

Split Before Lifting: A Candidate Level-I Break of QR-UOV

Branchwise base-field homotopy, scalar recombination, and Frobenius descent

Technical research note for integration into the structural-cryptanalysis manuscript
Authorship and final claims to be assigned after independent verification

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Status. This note records a candidate breakthrough found during a research takeover of the attached QR-UOV program. The mathematics has been stress-tested with exact toy eliminations and a machine-readable finite ledger, but it has not received independent review. The Boolean-gate numbers use the same explicit multiplication convention as the parent manuscript; they are diagnostics, not measured circuits.

Abstract

The parent manuscript’s projective direct forgery follows all $C = 2^a$ branches of a quadratic homotopy through one global geometric resolution over an auxiliary field. At Level I this gives a stress cost of $2^{151.639}$ Boolean gates, above the 2^{143} target. We show that the expensive global lift can be replaced by independent Hensel lifts of the C explicit start roots over the base field $\mathbb{F}_{127}$. Because the system is quadratic and every branch starts at a simple root, the coefficient recurrence uses the same constant inverse Jacobian at every precision and only $a(a+1)/2$ online power-series products.

The branches are combined only after lifting. A product tree reconstructs the separator polynomial and three scalar interpolation numerators: two extension-valued linear sketches for a Frobenius descent test, and the valuation-safe rational filter $H_0/(c + \lambda(x))$ that kills the hidden oil component. Full coordinates are reconstructed only for the few surviving roots. Under the parent manuscript’s gate model, the retry-aware Level-I total is $2^{141.126}$ gates, a margin of 1.874 bits below the category target and a 10.513-bit improvement over the parent ledger. The same transformation lowers the Level-III and Level-V diagnostics to $2^{191.630}$ and $2^{247.637}$ gates.

1 Executive result

Let $q = 127$. In the projective direct attack, the affine solver core consists of

$$a = m - 4$$

quadratic equations in a variables. Set

$$C = 2^a, \quad C_0 = 2^{a-1}(a+2).$$

Here C is the Bezout bound for the target fiber and C_0 is the degree bound for the one-parameter homotopy curve inherited from Safey El Din–Schost [1].

Theorem 1.1 (Candidate finite attack bound). *Assume the idealized public-random model and the Boolean-gate accounting conventions of the parent QR-UOV manuscript. Replace its global geometric-resolution lift by the branchwise algorithm in sections 3 and 4. Use the valuation-safe oil filter in section 5, the Frobenius descent test in section 6, and exact public verification. Then the expected Level-I work is bounded by*

$$2^{141.125734}$$

Boolean gates in the audit supplied with this note. The category target is 2^{143} , leaving a margin of 1.874266 bits.

Table 1: Headline comparison under the parent manuscript’s stress model. A positive margin means the attack lies below the category target.

Level	Parent cost	New cost	Target	New margin
I	151.639	141.126	143	+1.874
III	203.507	191.630	207	+15.370
V	260.389	247.637	272	+24.363

The Level-I conclusion is the new qualitative result. The parent manuscript already placed Levels III and V below their category targets; the new algorithm adds substantial slack there.

2 Inherited homotopy and degree bounds

Write the target square system as

$$f = (f_1, \dots, f_a) \in \mathbb{F}_q[X_1, \dots, X_a]^a,$$

with every f_i of degree at most two. The parent manuscript supplies a Vandermonde product start

$$g_i(X) = L_{\alpha_{2i-1}}(X)L_{\alpha_{2i}}(X), \quad 1 \leq i \leq a,$$

with $2a < q$, exactly $C = 2^a$ known roots $\xi_s \in \mathbb{F}_q^a$, and invertible $J_g(\xi_s)$ at every root. Consider

$$H(t, X) = (1 - t)g(X) + tf(X).$$

After saturation by the X -Jacobian, the homotopy curve has degree at most C_0 . For a well-separating linear form λ , its geometric resolution

$$q(t, U), \quad v_1(t, U), \dots, v_a(t, U)$$

has coefficient numerators and denominators of t -degree at most C_0 [1, Lemmas 8–10]. The target-specialization argument handles branches escaping to infinity and positive-dimensional target components [1, Lemma 12].

The parent algorithm lifts the entire resolution globally. The only change here is to split the simple start fiber into its C points before lifting, a standard operation in lifting-fiber implementations [5]. Symmetric recombination is deferred until terminal precision.

3 Independent base-field branch lifting

Fix one start root $\xi \in \mathbb{F}_q^a$. Hensel's lemma gives a unique formal branch

$$x(t) = \xi + y(t) \in \mathbb{F}_q[[t]]^a, \quad y(0) = 0, \quad H(t, x(t)) = 0.$$

The decisive observation is that no series-valued Jacobian inversion is needed.

Lemma 3.1 (Constant-Jacobian recurrence). *For every $n \geq 1$, the coefficient $y_n = [t^n]y(t)$ is the unique solution of*

$$J_g(\xi)y_n = -R_n,$$

where R_n depends only on $y_1, \dots, y_{n-1}$. In particular, the same constant matrix $J_g(\xi)^{-1}$ is used at every precision.

Proof. Write the degree-two Taylor expansions at ξ as

$$g(\xi + y) = J_g(\xi)y + Q_g(y)$$

and

$$(f - g)(\xi + y) = c + L(y) + Q_{f-g}(y).$$

Then

$$H(t, \xi + y) = J_g(\xi)y + Q_g(y) + t(c + L(y) + Q_{f-g}(y)).$$

Because $y(0) = 0$, the coefficient of t^n in every quadratic monomial $y_i y_j$ uses only coefficients of index strictly below n . The extra factor t makes the coefficient- n contribution of $L(y)$ depend only on y_{n-1} . Thus the only occurrence of y_n is $J_g(\xi)y_n$. $\square$

Let

$$h = \frac{a(a+1)}{2}.$$

Maintaining the h symmetric products $y_i y_j$ with an online multiplication algorithm computes a branch through precision P in

$$hR(P) + O(ahP + a^2P)$$

base-field operations, where $R(P)$ is the cost of one relaxed product. Online multiplication is a standard tool precisely for recurrences in which the next output coefficient is needed before the next input coefficient is known [3]. The audit uses the deliberately loose explicit envelope

$$R(P) \leq 2 \sum_{j=0}^{\lceil \log_2 P \rceil} \left\lceil \frac{P}{2^j} \right\rceil M(2^j) + 8P(\lceil \log_2 P \rceil + 1), \quad (1)$$

which is 54.224M(P) at Level I. This is much larger than the sharp constants known for online multiplication, and therefore acts as a substantial cushion rather than an optimization.

Remark 3.2 (Why this removes the cubic matrix factor). The global lift works in a degree- C coefficient algebra and repeatedly solves or inverts an $a \times a$ series matrix, leading to an a^3 outer factor in the parent ledger. Splitting first converts that operation into multiplication by a precomputed constant inverse for each branch, at $O(a^2P)$ per branch. The dominant nonlinear work is only the $h = O(a^2)$ online products.

4 Terminal scalar recombination

Let $x_s(t)$ be the C lifted branches and choose a well-separating

$$\lambda \in K'^a, \quad K' = \mathbb{F}_{q^d}.$$

Put $u_s(t) = \lambda \cdot x_s(t)$. The separator polynomial is

$$q(t, U) = \prod_{s=1}^C (U - u_s(t)). \quad (2)$$

For any scalar branch function $b_s(t)$, define its interpolation numerator

$$v_b(t, U) = \sum_{s=1}^C b_s(t) \prod_{r \neq s} (U - u_r(t)). \quad (3)$$

At a simple separated point, $b_s = v_b/q_U$. Both (2) and (3) are computed by one product tree. If a node stores (q_L, v_L) and (q_R, v_R) , its parent is

$$q = q_L q_R, \quad v = v_L q_R + q_L v_R. \quad (4)$$

For b scalar families, the tree uses one long product channel for q and two per numerator. With a collision-free Kronecker flattening and

$$N = 4CP,$$

all products at one tree level cost at most $\frac{1}{2}M(N)$ per channel, because $M(n)/n$ is increasing. Hence all $a = \log_2 C$ levels use at most

$$\frac{a}{2}(1 + 2b)M(N). \quad (5)$$

The new attack needs only $b = 3$ scalar families.

The rational functions in t are recovered coefficientwise. A general polynomial half-gcd now has asymptotic constant $17/4$ in front of $M(n) \log_2 n$ [4]; the audit uses constant 8, and includes sensitivity values through constant 128.

5 A valuation-safe oil-component filter

A direct use of the quadratic H_0 is incorrect. If a branch escaping to infinity has minimum coordinate valuation $\mu < 0$, then

$$\text{ord } H_0(x(t)) \geq 2\mu,$$

which can be smaller than the valuation μ needed in the Safey El Din–Schost specialization proof. This was caught during the correctness pass.

Choose $c \in K'$ and define

$$\psi(x) = \frac{H_0(x)}{c + \lambda(x)}. \quad (6)$$

Choose c so that the denominator is nonzero on every start point and every relevant finite target specialization. The forbidden offsets are $-\lambda(P)$, not $\lambda(P)$; there are at most $2C$ of them, so a uniform $c \in K'$ fails with probability at most $2C/q^d$.

The next lemma is the scalar version of the valuation argument in Safey El Din–Schost Lemma 12. It makes explicit that their coordinate numerators can be replaced by any scalar branch function having the same lower valuation bound.

Lemma 5.1 (Scalar specialization criterion). *Let $\Phi_1, \dots, \Phi_s$ be the lifted branches, let*

$$\mu_i = \min_j \text{ord } \Phi_{i,j}, \quad q = \prod_{i=1}^s (U - \lambda(\Phi_i)),$$

and let $b_i \in K'((t))$. Suppose that λ is well separating and that

$$\text{ord } b_i \geq \mu_i$$

for every branch with $\mu_i < 0$. Define

$$v_b = \sum_{i=1}^s b_i \prod_{j \neq i} (U - \lambda(\Phi_j)).$$

Then $\text{ord } v_b \geq \text{ord } q$. After multiplying q and v_b by the common normalizing power $t^{-\text{ord } q}$ and specializing at the target, the eliminant retains only the finite branch limits. An infinite-branch numerator summand may survive, but every such surviving summand is divisible by the full finite eliminant and hence vanishes at each finite simple root. Consequently a finite simple branch contributes its specialized scalar value through $b_i = v_b/q_U$.

Proof. For the term indexed by i , well separation gives

$$\text{ord} \left(b_i \prod_{j \neq i} (U - \lambda(\Phi_j)) \right) \geq \text{ord } b_i + \sum_{j \neq i, \mu_j < 0} \mu_j.$$

If $\mu_i < 0$, the hypothesis makes the right-hand side at least $\sum_{\mu_j < 0} \mu_j = \text{ord } q$. If $\mu_i \geq 0$, then b_i is regular at the target in the applications below, and omitting the i th index does not remove any negative summand, so the same inequality holds. Hence $\text{ord } v_b \geq \text{ord } q$ term by term. Multiplication by $t^{-\text{ord } q}$ gives integral coefficient series. Let S_∞ and S_0 denote the infinite and finite branches. At $t = 0$, the normalized eliminant is a nonzero scalar times $\prod_{s \in S_0} (U - \lambda(P_s))$. A finite-branch numerator summand specializes to the corresponding ordinary interpolation summand. An infinite-branch summand can survive normalization, but it contains every finite factor and is therefore divisible by the full finite eliminant. Reducing the specialized numerator modulo that eliminant, or merely evaluating at a finite simple root, removes all such terms. This is the scalar extension of the argument of Lemma 12; the accompanying local-separator note includes an explicit toy exhibiting the nonvanishing-but-divisible case. $\square$

Lemma 5.2 (Valuation bound). *Assume λ is well separating. For every branch Φ_i escaping to infinity, let*

$$\mu_i = \min_j \text{ord } \Phi_{i,j} < 0.$$

Then

$$\text{ord } \psi(\Phi_i) \geq \mu_i.$$

Consequently v_ψ satisfies theorem 5.1.

Proof. Well separation gives $\text{ord } \lambda(\Phi_i) = \mu_i$. Since $\mu_i < 0$, adding the constant c does not change the valuation:

$$\text{ord}(c + \lambda(\Phi_i)) = \mu_i.$$

Because H_0 has degree at most two,

$$\text{ord } H_0(\Phi_i) \geq 2\mu_i.$$

Thus $\text{ord } \psi(\Phi_i) \geq \mu_i$, and [theorem 5.1](#) applies. $\square$

Lemma 5.3 (Degree of the rational-filter graph). *Let $\Gamma \subset \mathbb{A}^{1+a}$ be the saturated homotopy curve and suppose $\deg \Gamma \leq C_0$. The closure of the graph of (6), saturated by $c + \lambda(X)$, has degree at most $2C_0$ in the standard affine embedding.*

Proof. Let $\bar{\Gamma} \subset \mathbb{P}^{a+1}$ have homogeneous coordinates $[Z : T : X_1 : \dots : X_a]$. Put

$$D_h = cZ + \lambda(X), \quad N_h = Z^2 H_0(X/Z),$$

so that D_h and N_h have degrees one and two. Away from $D_h = 0$, the graph is the image of

$$[Z : T : X] \mapsto [ZD_h : TD_h : X_1 D_h : \dots : X_a D_h : N_h].$$

All coordinates of this map are quadratic forms. The projection of its image back to $[Z : T : X]$ is generically one-to-one, because it is the graph map. Therefore a generic hyperplane section pulls back to a quadric section of $\bar{\Gamma}$, and the image degree is at most $2 \deg \bar{\Gamma} \leq 2C_0$. Saturating by D_h removes only the base-locus components and cannot increase this degree. $\square$

By the standard degree bounds for parametric geometric resolutions, the coefficient numerators and denominators associated with this graph have t -degree at most $2C_0$ [\[2\]](#). Hence

$$P = 4C_0 + 1 \tag{7}$$

Taylor coefficients suffice for rational reconstruction. At a finite candidate with separator value τ , recover

$$H_0(P) = \psi(P)(c + \tau).$$

The oil component has $H_0 = 0$ and is discarded. Before any coordinate recovery, require the recovered value to lie in $\mathbb{F}_q$ and to be a nonzero square.

6 Two-scalar Frobenius descent

The parent attack needs only base-field points, but the separator polynomial is factored over K' . Nine base-field sketches would reduce the probability of a non-base-field point surviving to q^{-9} . Two extension-valued sketches are much stronger.

Choose $\rho \leftarrow (K')^a$ uniformly and interpolate the two linear functions

$$b_0(P) = \rho \cdot P, \quad b_1(P) = \rho^{q^{d-1}} \cdot P,$$

where Frobenius is applied coefficientwise to ρ .

Lemma 6.1 (Frobenius descent test). *For every $P \in (K')^a$,*

$$b_1(P)^q - b_0(P) = \rho \cdot (P^q - P). \quad (8)$$

If $P \in \mathbb{F}_q^a$, the expression is zero for every ρ . If $P \notin \mathbb{F}_q^a$, a uniform ρ makes it zero with probability exactly q^{-d} .

Proof. Since $z^{q^d} = z$ for $z \in K'$,

$$b_1(P)^q = (\rho^{q^{d-1}} \cdot P)^q = \rho \cdot P^q.$$

This proves (8). If $P^q - P \neq 0$, the map

$$\rho \mapsto \rho \cdot (P^q - P)$$

is a nonzero K' -linear functional from $(K')^a$ onto K' , whose kernel has relative size $1/|K'| = q^{-d}$. $\square$

On the good-separator event, a linear factor $U - \tau$ over K' corresponds to a K' -rational point: Frobenius fixes τ , and separation forces it to fix the point. There are at most C represented points, so one random ρ eliminates every non-base-field point except with probability at most

$$\frac{C}{q^d}. \quad (9)$$

At Level I this is below 2^{-54} . The supplied exhaustive toy check over $\mathbb{F}_{25}$ finds exactly 25 passing ρ values among 625 for a fixed non-base-field point, and all 625 pass for a base-field point.

7 Root probability, abort cap, and exact acceptance

Let Z count base-field projective roots that pass the H_0 nonzero-square condition, without imposing nonsingularity. In the parent public-random model, linearity of expectation gives the exact bound

$$\mathbb{E}[Z] = \frac{q^r - 1}{q - 1} q^{-(r-1)} \frac{q - 1}{2q} = \frac{q^r - 1}{2q^r} < \frac{1}{2}.$$

Indeed, each projective point satisfies the first $r - 1$ independent random quadrics with probability $q^{-(r-1)}$, and the independent value of H_0 is a nonzero square with probability $(q - 1)/(2q)$. Thus, for an abort cap B ,

$$\Pr[Z > B] \leq \frac{1}{2(B + 1)}. \quad (10)$$

The audit uses $B = 2^{16}$. The Frobenius-filter failure in (9) is so small that its contribution to expected coordinate recoveries is negligible:

$$\mathbb{E}[\text{candidates per trial}] \leq \frac{1}{2} + B \frac{C}{q^d}.$$

The nonsingular eligible-root probability is sharpened slightly relative to the parent manuscript. Put

$$\mu = 1 - q^{-1}, \quad \pi = \prod_{j=1}^{r-1} (1 - q^{-j}).$$

Let M count all off-oil affine normalized roots and N the nonsingular ones. The parent second-moment calculation gives

$$\mathbb{E}[M] = \mu, \quad \mathbb{E}[M^2] = 2\mu + \mu^2 - (q-1)\mu q^{-r}, \quad \mathbb{E}[N] = \mu\pi.$$

Roots occur in pairs $z, -z$. The second Bonferroni inequality on the induced projective-point events, with nonsingular pairs bounded by all root pairs, gives

$$\begin{aligned} p_{\text{root}} &\geq \frac{\mathbb{E}[N]}{2} - \mathbb{E}\left[\binom{M/2}{2}\right] \\ &= \frac{\mu\pi}{2} - \frac{\mu^2}{8} + \frac{(q-1)\mu q^{-r}}{8} = 0.3690869822\dots \end{aligned} \tag{11}$$

The displayed expression is increasing for the actual $\mu \in [1 - q^{-1}, 1]$, so substituting $\mu = 1 - q^{-1}$ is valid. The subtraction in (10), the random affine chart, the separator event, the denominator event, and the Frobenius-filter event give a Level-I restart factor of

$$2.830442\dots$$

Acceptance remains zero error. Every surviving scalar candidate is expanded to full coordinates on demand. The algorithm then checks all unrecombined equations, the Jacobian condition, the target scaling, and the unchanged public map. Bad random choices can cause a restart or extra candidates, but cannot produce an accepted false forgery.

8 On-demand coordinate recovery

Globally interpolating all a coordinate numerators would multiply the terminal product-tree cost by roughly a . Instead, after scalar filtering leaves a separator value τ , re-run the branch lifts in streaming mode and evaluate the scalar product tree at $U = \tau$. This reconstructs

$$v_i(t, \tau) = \sum_s x_{s,i}(t) \prod_{r \neq s} (\tau - u_r(t))$$

for $1 \leq i \leq a$. Apply the same valuation normalization determined from q , specialize at $t = 1$, and divide by the derivative of the squarefree target separator polynomial. On a good separated simple root this yields the full point.

The expected number of such re-lifts is at most

$$\frac{1}{2} + B \frac{C}{q^d},$$

per outer trial. This also permits streaming memory: the first pass need not store all aCP coordinate coefficients.

9 Finite gate ledger

The audit retains the parent work unit

$$G_{\text{mul}}(b) = 2b^2 + b$$

for multiplication of b -bit field elements and its sparse irreducible fields of degrees 15, 22, 30. A long polynomial product over $\mathbb{F}_{q^d}$ is evaluated at $2d - 1$ distinct base-field points, followed by $2d - 1$ base polynomial products, dense interpolation in the extension generator, and sparse reduction. This is valid for every official profile because $2d - 1 < 127$.

Table 2: Level-I component costs. The total is the logarithm of the sum, not the maximum.

Component	$\log_2$ gates
Initial base-field branch lifts	139.361
Valuation-safe filter series	133.778
Terminal scalar recombination	140.268
On-demand coordinate re-lifts	138.361
Extension-valued leaf formation	129.785
Factorization, filtering, and verification envelope	115.710
Preprocessing	63.368
Total	141.126

The terminal tree has three value families and therefore

$$\frac{a}{2}(1 + 2 \cdot 3) = 175$$

long-product units at Level I. Rational reconstruction contributes 124.357 units, expected coordinate candidates contribute 5.671 units, and the audit adds 64 miscellaneous units, for 369.029 units per trial before retries.

Table 3: Sensitivity to the constant multiplying $M(n) \log n$ in rational reconstruction. The main ledger uses 8, versus the published asymptotic general half-gcd constant $17/4$.

Constant	8	10	16	32	64	128
Level-I total	141.126	141.191	141.372	141.765	142.329	143.048

Thus the attack remains below 2^{143} even when the rational-reconstruction constant is increased from 8 to 64. A constant of 128 would erase the finite Level-I margin. This is one reason the result should be presented as a carefully audited model-based break, not as a measured implementation claim.

10 Exact toy validations

The bundle contains three algebraic toy checks and a separately written ledger verifier.

1. `verify_split_lift_ledger.py`: independently reimplements the complete finite ledger without importing either audit module. It reproduces all three totals to better than $5 \cdot 10^{-10}$ bits.
2. `toy_branch_recombine.py`: independently lifts all branches of a two-variable Boolean product start and verifies coefficient-for-coefficient equality with the direct resultant. It also verifies the derivative coordinate identity.
3. `toy_vandermonde_branch_recombine.py`: repeats the test for the Vandermonde start used by the attack and verifies coordinate, random-linear, and quadratic-value interpolation numerators.

4. `toy_frobenius_descent.py`: exhaustively checks [theorem 6.1](#) in $\mathbb{F}_{25}^2$. A fixed non-base-field point passes for 25 of 625 random functionals, exactly $1/|\mathbb{F}_{25}|$; a base-field point passes for all 625.

The ledger cross-check protects against shared-code arithmetic errors; the toys protect the algebraic identities on which the redesign depends. Neither substitutes for an expert proof review.

11 Remaining proof and engineering obligations

Before merging this note into the main paper, an independent reviewer should check the following points line by line.

1. **Graph and specialization audit.** Independently check the projective graph argument in [theorem 5.3](#) and the scalar extension of Safey El Din–Schost Lemma 12 in [theorems 5.1](#) and [5.2](#). These two points now have proofs in the note, but they are the most nonstandard algebraic steps.
2. **Online-product envelope.** Either include a full dyadic scheduling proof of (1) or replace it by a directly quoted theorem with explicit finite constants. The current factor 54.224 is intentionally loose, so this is unlikely to worsen the headline.
3. **Finite rational reconstruction.** The audit’s constant 8 is above the asymptotic $17/4$ half-gcd constant, and the sensitivity table remains below target through 64. A completely formal circuit theorem would still need explicit lower-order terms.
4. **Memory and locality.** The algorithm is streamable, but its absolute storage and data movement are astronomical. The parent security comparison counts Boolean work rather than a realistic memory hierarchy. This assumption must be displayed prominently.
5. **Seeded transfer.** The new solver changes no public-random or ideal-primitive reduction, so the parent manuscript’s seeded-transfer caveats remain unchanged. They should be inherited verbatim rather than silently strengthened.

12 Assessment

The route is qualitatively stronger than a constant-factor refinement of the old global lift. It exploits two structural facts that the parent ledger did not use together:

1. the start fiber is already completely split over $\mathbb{F}_{127}$, and
2. a quadratic branch recurrence can be advanced with a fixed base-field Jacobian and online products.

The extension field is postponed until terminal symmetric recombination, and full coordinate interpolation is postponed until after scalar descent and oil filtering. These changes remove the global a^3 series-matrix cost and most extension-field arithmetic. The resulting 10.5-bit Level-I improvement is large enough to cross the target with nearly two bits of audited slack.

A separately written ledger implementation now reproduces every headline value. The most important next action is therefore an external adversarial proof audit of [theorems 3.1](#), [5.1](#), [5.3](#) and [6.1](#), followed by integration into the main manuscript and, ideally, a prototype of the branch scheduler and terminal product tree.

References

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